BOTANY



Trial Event

MIT Science Olympiad Invitational
January 25, 2020

Names: _____

Team Name: _____ Team Number: ____

Overall Score: _____ Rank: ____

Time: 50 Minutes

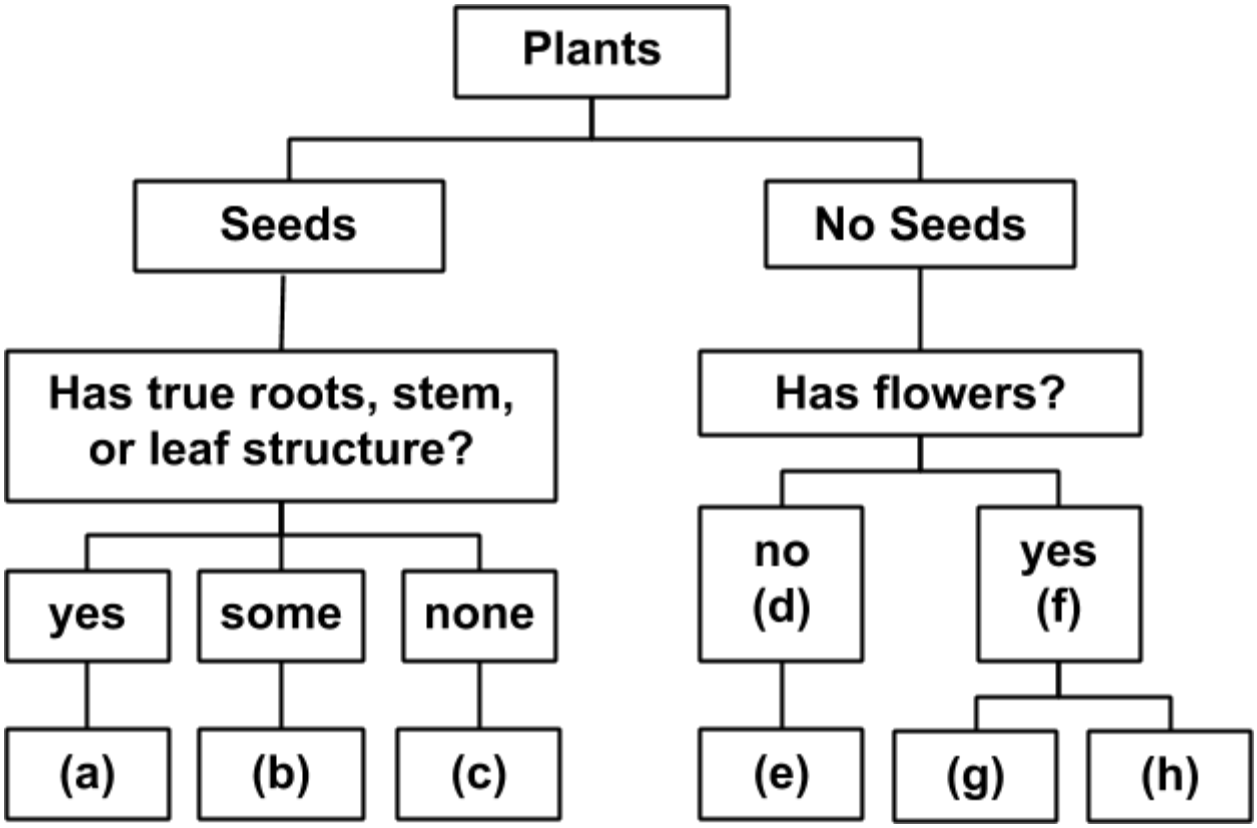
Number of Questions: 70

Number of Points: _____/200

Team Number:_____ Score:_____

Team Number:_____ Score:_____

Plant Classification (2 points per label)



a)_____

b)_____

c)_____

d)_____

e)_____

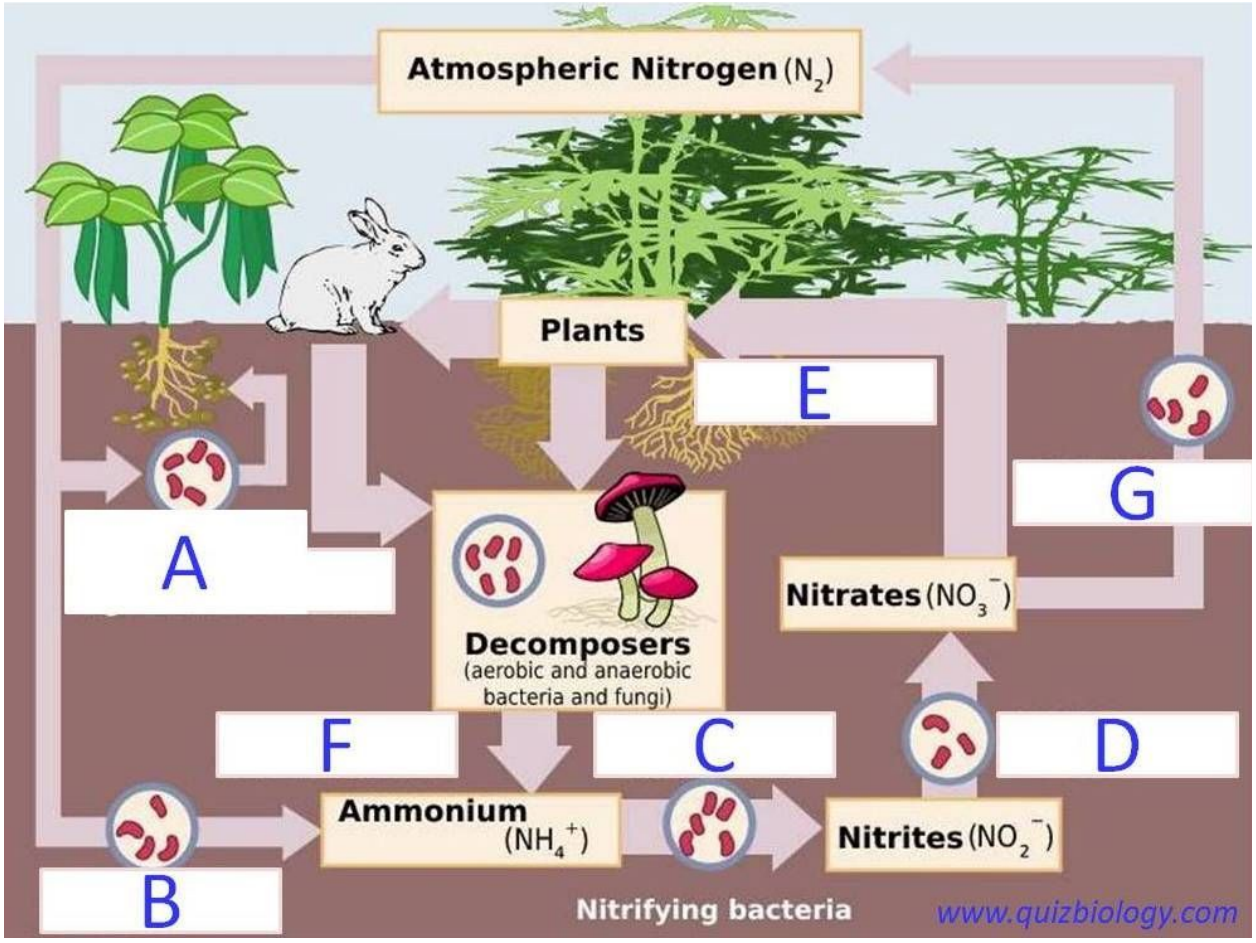
f)_____

g)_____

h)_____

Team Number:_____ Score:_____

The Nitrogen Cycle (2 points per label)



a)_____

b)_____

c)_____

d)_____

e)_____

f)_____

g)_____

Team Number:_____ Score:_____

Multiple Choice (2 points each)

_____ 1. Starch is a polymer of

A. glucose B. sucrose C. fructose D. cellulose

_____ 2. The carbohydrate produced during the Calvin Cycle is

A. starch B. glucose C. PGA D. triose phosphate (PGald)

(PGA = phosphoglyceric acid; PGald = phosphoglyceraldehyde)

Use the following answer choices for questions 3-7.

A. glycolysis B. Krebs cycle C. electron transport chain (ETC) D. A and B

_____ 3. O2 (oxygen) is reduced to H2O (water)

_____ 4. pyruvic acid —>—>—> 3 CO2

_____ 5. a proton (H+) gradient forms and ATP is produced

_____ 6. glucose —>—>—> pyruvic acid

_____ 7. electrons are removed from NADH and FADH2

_____ 8. During photosynthesis, oxygen (O2) is released from

A. carbon dioxide (CO2) during the light reactions

B. carbon dioxide (CO2) during the Calvin cycle

C. water (H2O) during the light reactions

D. water (H2O) during the Calvin cycle

_____ 9. Ploidy refers to

A. homologous chromosomes B. the number of individual chromosomes in a nucleus

C. the number of sets of chromosomes in a nucleus

_____ 10. The phenomenon of double fertilization involves the fusion between

A. the egg and two sperm

B. the egg and one sperm and the polar nuclei (central cell) and the other sperm

C. the egg and one sperm and the synergids and the other sperm

D. the egg and one sperm and the antipodals and the other sperm

_____ 11. A spore

A. becomes a gametophyte by mitosis B. becomes a gametophyte by meiosis

C. becomes a sporophyte by mitosis D. becomes a sporophyte by meiosis E. is a gamete

____ 12. A gametophyte produces

A. spores by mitosis B. spores by meiosis C. gametes by mitosis D. gametes by meiosis

____ 13. A sporophyte produces

A. spores by mitosis B. spores by meiosis C. gametes by mitosis D. gametes by meiosis

Team Number:_____ Score:_____

_____ 14. The Calvin cycle of photosynthesis
A. occurs in the thylakoids and results in the production of ATP and NADPH
B. occurs in the stroma and results in the production of ATP and NADPH
C. occurs in the thylakoids and results in the production of carbohydrate (sugar)
D. occurs in the stroma and results in the production of carbohydrate (sugar)

_____ 15. The light reactions of photosynthesis
A. occur in the thylakoids and result in the production of ATP and NADPH
B. occur in the stroma and result in the production of ATP and NADPH
C. occur in the thylakoids and result in the production of carbohydrate (sugar)
D. occur in the stroma and result in the production of carbohydrate (sugar)

_____ 16. If a secondary cell wall is produced, it is often strengthened by the addition of
A. cellulose B. cutin C. calcium pectate (pectin) D. lignin

_____ 17. Plant cells that are usually dead at maturity include all of the following except
A. companion cells B. sclereids C. vessel elements D. tracheids

_____ 18. The tissue that moves dissolved sucrose in a generally downward direction is
A. xylem B. phloem C. tracheid D. vessel E. C and D

_____ 19. The structure that surrounds the cell and controls the movement of materials into and out of the cell is the
A. cell wall B. plasma membrane C. middle lamella D. plasmodesmata

_____ 20. A plastid specialized for the storage of starch is the
A. amyloplast B. chloroplast C. chromoplast

_____ 21. The light colored, conducting xylem is called
A. hardwood B. summer wood C. heartwood D. sapwood

_____ 22. The tissue that divides the wood and the bark is the
A. cork B. periderm C. vascular cambium D. cork cambium E. secondary phloem

_____ 23. Lateral roots are produced by the
A. pericycle B. root hairs C. buds D. xylem E. endodermis

_____ 24. Wood is
A. secondary phloem B. secondary xylem C. A and B D. secondary xylem and bark

Team Number:_____ Score:_____

_____ 25. The dark, nonfunctional xylem at the center of a tree is called
A. hardwood B. late wood C. heartwood D. sapwood E. early wood

_____ 26. Growth rings can be seen in cross sections of trees because of differences in the sizes of the cells of the
A. sapwood and heartwood B. early wood and late wood
C. primary and secondary xylem D. primary and secondary phloem

_____ 27. Bark is a nontechnical name for
A. the cork B. the periderm C. all of the tissues outside the vascular cambium
D. all of the tissues outside the cork cambium

_____ 28. A nail is pounded into the side of a tree, 5 feet above the ground. The tree grows 1 foot per year. How far above the ground will the wire be after 3 years?
A. 5 feet B. 6 feet C. 7 feet D. 8 feet E. the tree will be dead by then

_____ 29. Loose collections of cells that allow gas exchange are called
A. lenticels B. stomata C. bud scales D. leaf scars

_____ 30. The roots of some plants are modified to parasitize other plants. These roots are called
A. mycorrhizae B. haustoria C. adventitious D. radicles

_____ 31. The poisonous nightshade plant is the ancestor of
A. Tomatoes B. Potatoes C. Chili and Bell Peppers D. All of the Above

_____ 32. Nitrifying bacteria ultimately creates
A. Nitrates B. Nitrites C. Ammonia D. Ammonium

_____ 33. Unlike most plants, these structures in cacti open at night, allowing them to thrive in hot and dry climates
A. lenticels B. stomata C. bud scales D. leaf scars

_____ 34. The Mendel pea plant experiment resulted in
A. genetic diversity of pea plants B. Punnett squares C. fundamental laws of inheritance

_____ 35. The plant type that produces the greatest percentage of atmospheric oxygen is
A. trees B. algae (phytoplankton) C. flowers D. succulents and cacti

Team Number:_____ Score:_____

Propagating Produce (5 points each)

Questions 1-8.

You are standing in the produce section of your local supermarket, surveying the local selection. You want to start a small garden from kitchen scraps left over from cooking with the produce, rather than starting from seed. Indicate whether or not it would be possible to propagate a plant from a piece of produce for each plant type to produce a similar edible plant and reasoning why.

1. Carrots
2. Cavendish Banana
3. Scallions/Green Onions (with roots)
4. Mango fruit
5. Avocado
6. Pineapple
7. Lettuce/Cabbage
8. Lemon

9. You notice a potato with green sprouts coming from its eyes. Is it still safe to eat or a potential poison? Use reasoning from the growth stages of tubers.
10. Oh no! Your lettuce/leafy green vegetable has wilted. Should you attempt to dry it in a salad spinner or soak it in water? Explain why for full credit.

Team Number:_____ Score:_____

Biodiversity (5 points each)

1. Artificial banana flavoring does not taste like the Cavendish bananas we enjoy as fresh fruit today. Rather, it is the chemical isoamyl acetate associated with the flavor of the Gros Michel cultivar, now extinct due to fungal plagues such as Panama disease. Explain the importance of biodiversity in protecting our current banana crop for generations to come.

2. Potatoes were once seen as poisonous and unfit for human consumption. Explain how human intervention led to the dispersion and adoption of the potato.

3. Why is crop rotation important in the production of corn, soybeans, wheat, and/or sunflowers?

4. Genetically modified organisms (GMOs) are often a source of public scrutiny and skepticism. Provide examples of how GMO plants have positively and negatively affected agriculture.

5. Describe plant succession following a wildfire. What plants should be introduced on rocks and soil to revitalize a burned area?

Hydroculture and Hydroponics (5 points each)

Tired of never knowing when it's right to water your plants (or overwatering them :( ) you decide to look into hydroculture and hydroponics, the process of growing plants without soil and with large amounts of water.

1. How is it possible for plants to be grown without soil? Explain using essential elements of plant growth and development.

2. What types of nutrient supplements would be needed for hydroponic plants to thrive?

3. Describe existing use of hydroculture in commercial farming (plants that are grown at scale while partially or entirely submerged in water).

4. Explain whether or not fish are beneficial for a hydroponic system.

5. Despite a typical hot and dry climate of origin, some types of succulents and cacti can be grown in water when certain conditions are met. What are these conditions?

MIT Invitational - Repaired practice edition

Division C | Botany | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (4 points) Authored written substitute for unavailable source material: Distinguish hydroponics and aquaculture.

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

Answer: _____
_____
_____